

CURRICULUM VITAE
(last update: November 2025)
Bahman Angoshtari

University of Miami
Department of Mathematics
1365 Memorial Drive
Coral Gables, FL 33146

Email: bangoshtari@miami.edu
Web: <https://www.math.miami.edu/~bangoshtari/>
Office: Ungar 547 **Phone:** 305.284.3742
Citizenship: U.S.

RESEARCH INTERESTS

stochastic optimal control, financial and actuarial mathematics.

EMPLOYMENT

Assistant Professor, University of Miami, Coral Gables, USA, Aug. 2020 – present.

Postdoc, University of Washington, Seattle, USA, Aug. 2017 – Jun. 2020.

Postdoc, University of Michigan, Ann Arbor, USA, Sep. 2014 – Jul. 2017.

Postdoc, The Chinese University of Hong Kong, Hong Kong, Jan. 2014 – Aug. 2014.

EDUCATION

University of Oxford, UK, D.Phil. in Mathematics, Oct. 2009 – Feb. 2014.

Thesis: “Stochastic modeling and methods for portfolio management in cointegrated markets.”

Advisor: Professor Thaleia Zariphopoulou.

University of Twente, the Netherlands, M.Sc. in Applied Mathematics, Sep. 2007 – Aug. 2009.

Sharif University of Technology, Iran, B.Sc. in Industrial Engineering, Oct. 2000 – Aug. 2004.

PUBLICATIONS

Under review:

“Optimal consumption under loss-averse multiplicative habit-formation preferences,” with X. Yu and F. Yuan, available at arXiv:2406.20063 (last update: April 2025).

Published:

11. Angoshtari, B., and S. Duan, “Rank-Dependent Predictable Forward Performance Processes,” *Probability, Uncertainty and Quantitative Risk*, 9(2):181–218, 2024.

10. Angoshtari, B., E. Bayraktar, and V. R. Young, “Optimal Consumption under a Habit-Formation Constraint: the Deterministic Case,” *SIAM Journal on Financial Mathematics*, 14(2):557–597, 2023.

9. Angoshtari, B., “Predictable forward performance processes in complete markets,” *Probability, Uncertainty and Quantitative Risk*, 8(2):141–176, 2023.

8. Angoshtari, B., E. Bayraktar, and V. R. Young, “Optimal Investment and Consumption under a Habit-Formation Constraint,” *SIAM Journal on Financial Mathematics*, 13(1):321–352, 2022.

7. Angoshtari, B., T. Zariphopoulou, and X. Zhou, “Predictable Forward Performance Processes: The Binomial Case,” *SIAM Journal on Control and Optimization*, 58(1):327–347, 2020.

6. Angoshtari, B. and T. Leung, “Optimal Trading of a Basket of Futures Contracts,” *Annals of Finance*, 16(2):253–280, 2020.

5. Angoshtari, B., E. Bayraktar, and V. R. Young, "Optimal Dividend Distribution Under Drawdown and Ratcheting Constraints on Dividend Rates," *SIAM Journal on Financial Mathematics*, 10(2):547–577, 2019.
4. Angoshtari, B. and T. Leung, "Optimal Dynamic Basis Trading," *Annals of Finance*, 15(3):307–335, 2019.
3. Angoshtari, B., E. Bayraktar, and V. R. Young, "Optimal investment to minimize the probability of drawdown," *Stochastics*, 88(6):946–958, 2016.
2. Angoshtari, B., E. Bayraktar, and V. R. Young, "Minimizing the probability of lifetime drawdown under constant consumption," *Insurance: Mathematics and Economics*, 69:210–223, 2016.
1. Angoshtari, B., E. Bayraktar, and V. R. Young, "Minimizing the expected lifetime spent in drawdown under proportional consumption," *Finance Research Letters*, 15:106–114, 2015.

Permanent working papers:

"Optimal Insurance to Minimize the Probability of Ruin: Inverse Survival Function Formulation," with V. R. Young, available at arXiv:2012.03798 (last update: Dec. 2020)

"On the market-neutrality of optimal pairs-trading strategies", available at arXiv:1608.08268 (last update: Aug. 2016)

Theses:

Angoshtari, B. (2013): "Stochastic modeling and methods for portfolio management in cointegrated markets", D.Phil. Thesis, University of Oxford, U.K.

Angoshtari, B. (2009): "On utility of wealth maximization". M.Sc. Thesis, University of Twente: The Netherlands.

GRANTS

Simons Foundation's Travel Support for Mathematicians, 2022–2027.

PhD STUDENTS

Shida Duan (August 2022–June 2025).

TALKS (invited and contributed)

2025 Mathematical Congress of the Americas, Miami, July 2025

SIAM Conference on Financial Mathematics and Engineering, Miami, July 2025

Byrne B2A2 (Back to Ann Arbor) Conference, University of Michigan, Ann Arbor, June 2025

Financial Mathematics Seminar, Worcester Polytechnic Institute, Worcester, April 2025

AMS Spring Eastern Sectional Meeting, Hartford, April 2025

2024 8th Eastern Conference on Mathematical Finance, University of Toronto, Toronto, Sep. 2024

1st INFORMS Conference on Financial Engineering and FinTech, Hong Kong Polytechnic University, Hong Kong, August 2024

12th Bachelier World Congress, Rio de Janeiro, July 2024

37th New England Statistics Symposium, University of Connecticut, Storrs, May 2024

- 2023** 7th Eastern Conference on Mathematical Finance, North Carolina State University, Raleigh, October 2023
- AMS Southeastern Sectional Meeting, Georgia Institute of Technology, Atlanta, March 2023
- Mathematical Finance, Control and Optimization seminars, University of Connecticut, Feb. 2023
- Joint Mathematics Meetings, Boston, January 2023
- 2022** Department of Mathematics Colloquium, Wichita State University, Oct. 2022
- 2021** Mathematical Finance Seminar (virtual), Florida State University, Tallahassee, Nov. 2021
- SIAM Conference on Financial Mathematics and Engineering (virtual), Jun. 2021
- AMS Spring Eastern Sectional Meeting (virtual), Mar. 2021
- Stochastic Finance Seminars (virtual), University of Warwick, Coventry, Feb. 2021
- 2020** Financial Mathematics Seminar (virtual), University of Michigan, Ann Arbor, Nov. 2020.
- 2019** Financial Mathematics Seminar, University of Michigan, Ann Arbor, Dec. 2019.
- Department of Mathematics Colloquium, University of Miami, Coral Gables, Dec. 2019
- Department of Mathematics Colloquium, California State University, Fullerton, Dec. 2019
- INFORMS annual meeting, Seattle, Oct. 2019.
- 2nd Biennial Meeting of SIAM Pacific Northwest Section, Seattle, Oct. 2019.
- Applied Mathematics: The Next 50 Years, University of Washington, Seattle, June 2019.
- 2018** The 9th Western Conference on Mathematical Finance, University of Southern California, Los Angeles, Nov. 2018.
- INFORMS annual meeting, Phoenix, Nov. 2018.
- SIAM annual meeting, Portland, July 2018.
- Department of Mathematics, Worcester Polytechnic Institute, Worcester, Mar. 2018.
- 2017** Department of Applied Mathematics CFRM Seminar, University of Washington, Seattle, Feb. 2017.
- Department of Math. and Stats. Colloquium, Georgetown University, Washington D.C., Jan. 2017.
- 2016** SIAM Conference on Financial Mathematics & Engineering, Austin, Nov. 2016.
- Financial/Actuarial Seminar, University of Michigan, Ann Arbor, Nov. 2016.
- AMS Spring Southeastern Sectional Meeting, University of Georgia, Mar. 2016.
- 2015** Financial/Actuarial Seminar, University of Michigan, Ann Arbor, Nov. 2015.
- 2014** Financial/Actuarial Seminar, University of Michigan, Ann Arbor, Sep. 2014.
- Applied Mathematics Colloquium, Illinois Institute of Technology, Chicago, Feb. 2014.
- The Chinese University of Hong Kong, Hong Kong, Jan. and Mar. 2014.

2012 The Oxford-Man Institute Internal Seminar, Oxford, Jan. 2012.

2011 The Man Investments Quant Forum, Oxford, Aug. 2011.

Mathematical and Computational Finance Group internal Seminar, Mathematical Institute, Oxford, Feb. 2011.

The Oxford-Man Institute Internal Seminar, Oxford, Feb. 2011.

TEACHING

University of Miami

MTH 721 - " <i>Mathematical Probability</i> "	Spring 25
MTH 721 - " <i>Stochastic Differential Equations (PhD topic course)</i> "	Fall 23
MTH 692&693 - " <i>Topics in Mathematics (ML in Finance)</i> "	Fall 20, 21
MTH 648 - " <i>Stochastic Calculus with Application to Finance</i> "	Spring 21, 25
MTH 647 - " <i>Introduction to Mathematical Finance</i> "	Fall 20, 23, 25
MTH 645 - " <i>Optimization Methods</i> "	Fall 20, 21, 24, 25
MTH 224 - " <i>Introduction to Probability and Statistics</i> "	Fall 21, 22, Spring 23, 24

University of Washington

CFRM 521 - " <i>Machine Learning for Finance</i> "	Spring 2018, 2019, 2020
CFRM 505 - " <i>Monte Carlo Methods in Finance</i> "	Winter 2020
CFRM 502 - " <i>Financial Data Science</i> "	Winter 2018, 2019, 2020
CFRM 420 - " <i>Introduction to Computational Finance and Financial Econometrics</i> "	Summer 2018
CFRM 410 - " <i>Probability and Statistics for Computational Finance</i> "	Winter 2019

University of Michigan

Math 526 - " <i>Discrete Space Stochastic Processes</i> "	F2014, W2015, W2016
Math 472 - " <i>Numerical Methods with Financial Applications</i> "	F2015
Math 425 - " <i>Introduction to Probability</i> "	S2017
Math 423 - " <i>Mathematics of Finance</i> "	F2016, W2017
Math 115 - " <i>Calculus I</i> "	F2014

SERVICE

- Co-organizer of the special session on "*Mathematical Finance and Risk Management*" at the AMS Spring Southeastern Sectional Meeting, Savannah (March 2026).
- Co-organizer of the special session "*Recent Advances in Mathematical Finance and Related Fields*" at the Mathematical Congress of the Americas 2025, Miami (July 2025).
- Co-organizer of the special session "*Recent Advances in Mathematical Finance and Related Fields*" at the Mathematical Congress of the Americas 2025, Miami (July 2025).
- Member of the organizing committee for the "*SIAM Conference on Financial Mathematics and Engineering*," Miami (July 2025). Also, co-organizer of the minisymposium "*Information Acquisition and Optimal Control Under Model Uncertainty*" at the conference.
- Organizer of the minisymposium "*Advances in stochastic control with financial applications*" at the 12th Bachelier World Congress, Rio de Janeiro (July 2024).
- Organizer of the "*Mathematical Finance Seminar*" at the department of Mathematics, University of Miami, Coral Gables, (Spring 2024, 2025).
- Member of the MSMF (M.Sc in Mathematical Finance) Steering Committee, University of Miami (August 2020–present).

- Organizer of the session “*Stochastic Control and Optimal Portfolio Choice*” in SIAM Annual Meeting, Portland (July 2018).

Journal reviewing: Annals of Finance; Applied Mathematics and Optimization; Applied Mathematical Finance; Decisions in Economics and Finance; Finance and Stochastics; Frontiers of Mathematical Finance; Insurance: Mathematics and Economics; Journal of Optimization Theory and Applications; Mathematics and Financial Economics; Mathematical Control and Related Fields; Mathematical Finance; Mathematical Methods of Operations Research; Mathematics of Operations Research; Operations Research Letters; Quantitative Finance; Scandinavian Actuarial Journal; SIAM Journal on Control and Optimization; SIAM Journal on Financial Mathematics; Stochastic Models.

AWARDS AND MEMBERSHIPS

Institute of Mathematical and Statistical Innovations, University of Chicago; long program visiting scholar, April-May 2022.

The Oxford-Man Institute scholarship and student membership, 2009 – 2013; associate membership, 2014 – Dec. 2016.

Award for the best M.Sc. thesis in the faculty of Electrical Engineering, Mathematics and Computer Science, University of Twente, 2010.

Huygens Scholarship, awarded by the Dutch Ministry of Education, Culture and Science (OCW), 2007 – 2009.

Valedictorian Graduate from the Department of Industrial Engineering, Sharif University of Technology, 2004.

NON-ACADEMIC EMPLOYMENT

Intern, The Oxford-Man Institute of Quantitative Finance, Oxford, UK, Spring 2013.

Intern, Man Research Laboratory, Oxford, UK, Summer 2012.